

Quenched Bond Disorder in Frustrated Non-Collinear Magnets

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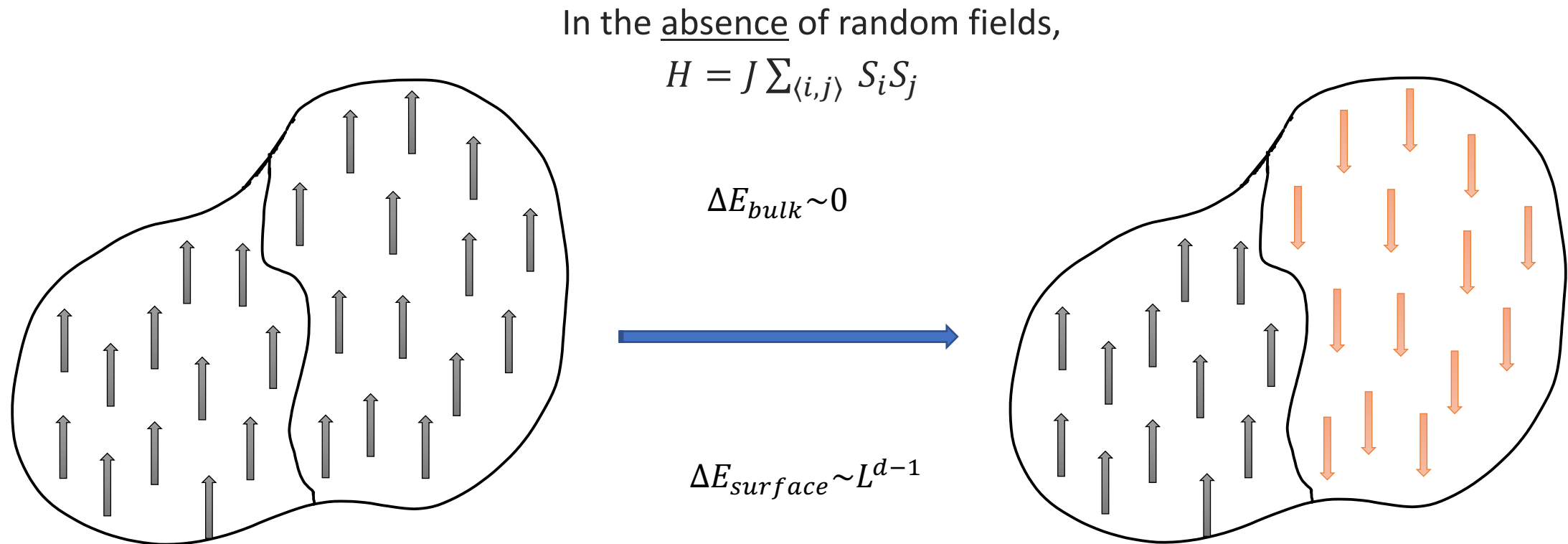
Quenched Disorder: Random Fields

Lower critical dimensions

Quenched Disorder: Random Fields

Random fields and the Imry-Ma criterion

- A simple picture with Ising spins:



Quenched Disorder: Random Fields

Random fields and the Imry-Ma criterion

- A simple picture with Ising spins:

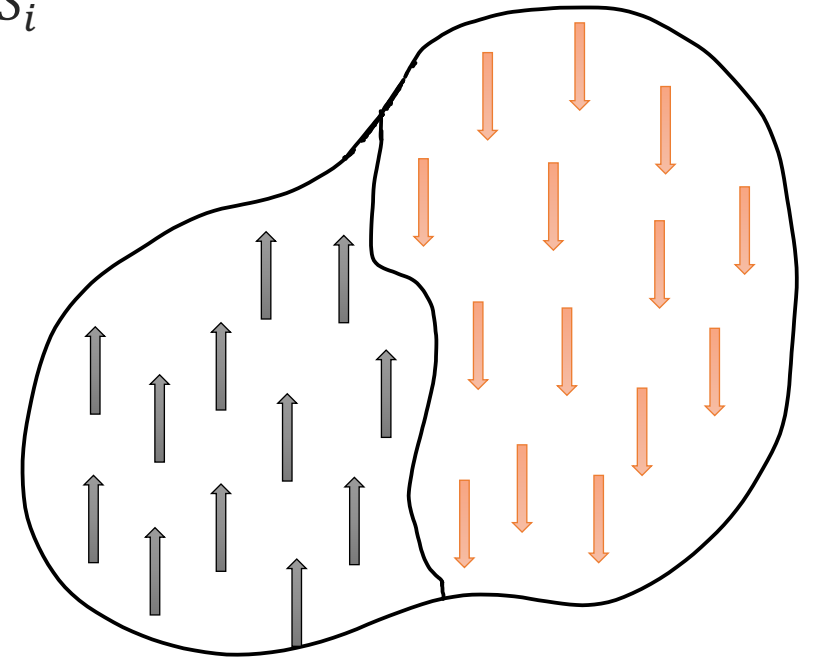
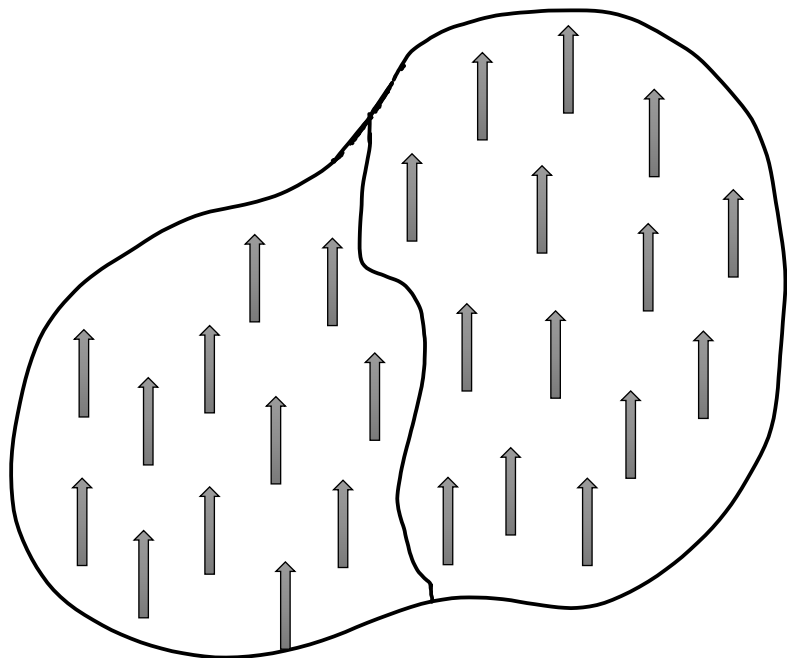
In the presence of random fields,

$$H = J \sum_{\langle i,j \rangle} S_i S_j + \sum_i h_i S_i$$

$$\Delta E_{bulk} \sim \sum_i h_i \sim L^{d/2}$$



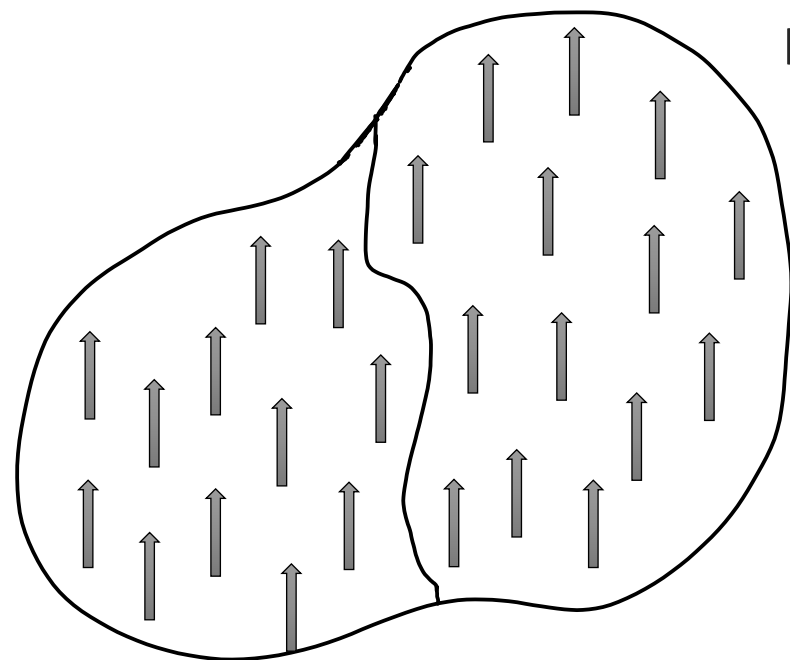
$$\Delta E_{surface} \sim L^{d-1}$$



Quenched Disorder: Random Fields

Random fields and the Imry-Ma criterion

- A simple picture with Ising spins:



In the presence of random fields,

$$H = J \sum_{\langle i,j \rangle} S_i S_j + \sum_i h_i S_i$$

long range order is destroyed if,

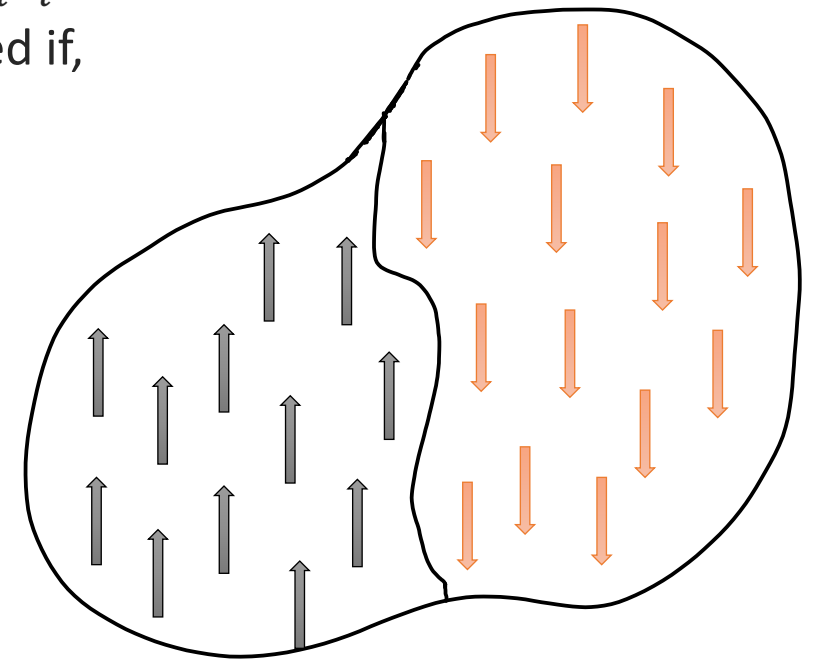
$$\Delta E_{bulk} \geq \Delta E_{surface}$$

$$\approx L^{\frac{d}{2}} \geq L^{d-1}$$

$$\approx d \leq 2$$

,the critical dimension
for discrete symmetries.

Favoured!



Quenched Disorder: Random Fields

Random fields and the Imry-Ma criterion

- For continuous symmetries:

A Heisenberg model for illustration,

$$H = J \sum_{\langle i,j \rangle} \vec{S}_i \cdot \vec{S}_j + \sum_i \vec{h}_i \cdot \vec{S}_i$$

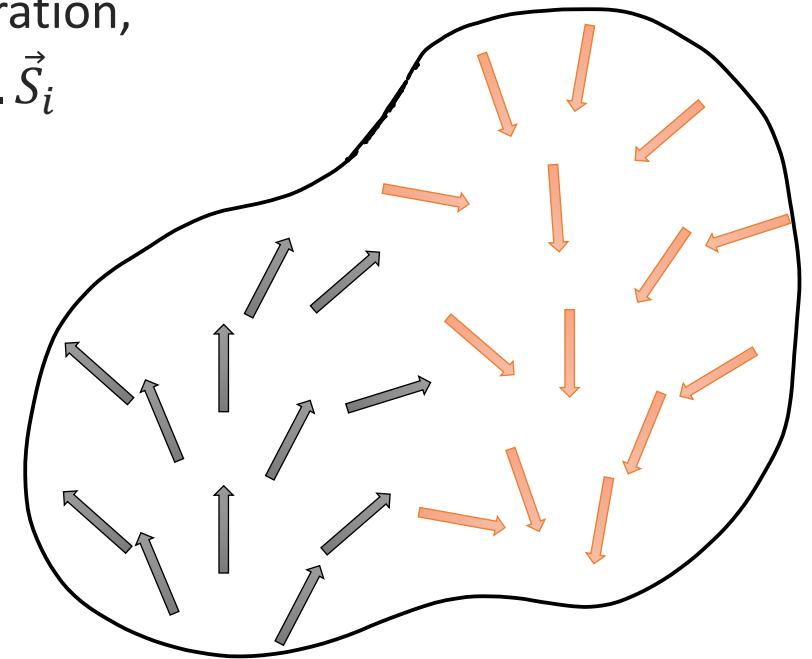
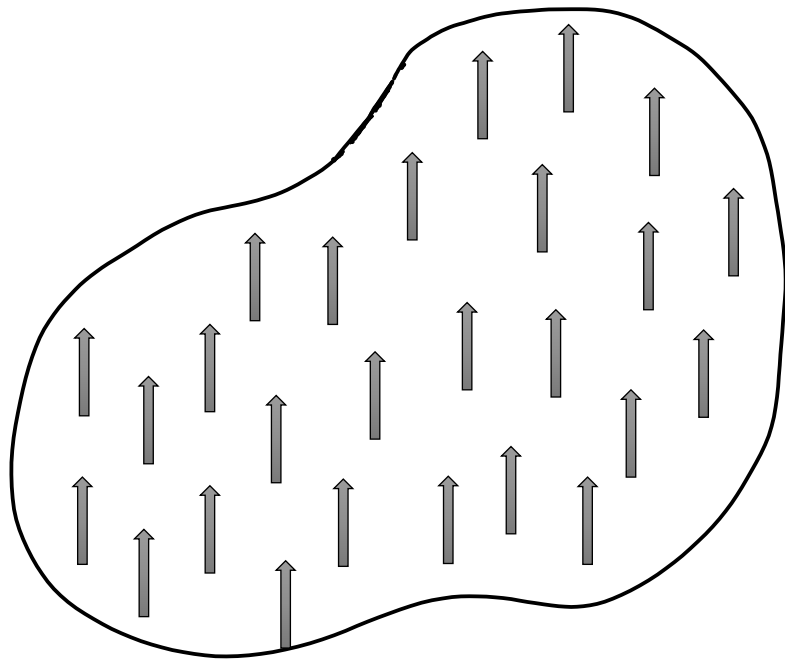
$$\Delta E_{bulk} \sim L^{d/2}$$

$$\Delta E_{surface} \sim L^{d-2}$$

So the critical dimension for continuous symmetries is,

$$\frac{d}{L^2} \geq L^{d-2} \text{ i.e. } d \leq 4$$

(Imry and Ma, PRL, 1975)



L

$$\Delta E_{surface} = \int d^d x \nabla \theta^2 \sim L^{d-2}$$

Quenched Disorder: Random Fields

Random fields and the infrared divergence

■ For continuous symmetries:

- A generic action couples the goldstone modes with the random fields.

$$S = \frac{1}{\beta} \sum_{\omega_n} \left(\int \frac{d^d k}{(2\pi)^d} \sum_a \psi_a(-\omega_n, -k) G^{-1}(\omega_n, k) \psi_a(\omega_n, k) + \beta \delta_{\omega_n, 0} \int \frac{d^d k}{(2\pi)^d} \sum_a h_a(-k) \psi_a(\omega_n, k) \right)$$

- The correlation function has a divergent contribution for even infinitesimal random field disorder, $\overline{h_a(x) h_b(x')} = h_0^2 \delta_{ab} \delta^{(d)}(x - x')$

$$\overline{C_a(x, x')} = \frac{1}{\beta} \sum_{\omega_n} \int \frac{d^d k}{(2\pi)^d} G(\omega_n, k) e^{-ik(x-x')} + h_0^2 \int \frac{d^d k}{(2\pi)^d} G^2(0, k) e^{-ik(x-x')}$$

Bare expectation value,

$$\int \frac{d^d k}{2\pi} \left(\frac{1}{k^2} \right)^2 \sim \text{infrared divergent for } d \leq 4$$

Hence, the lower critical dimension for random field systems with continuous symmetry is,

(Imry and Ma, PRL, 1975; Aharony, SSC, 1978)

$$d \leq 4$$

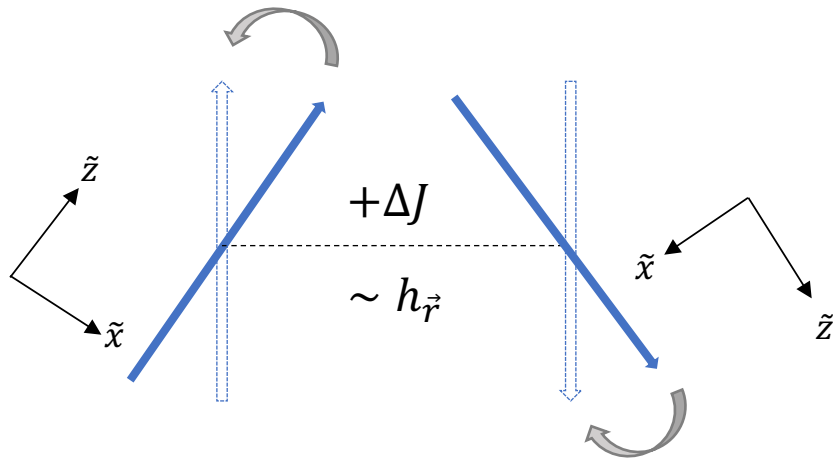
Quenched Disorder: Random Couplings

Lower critical dimension of disordered frustrated non-collinear magnets:
Infrared divergence

Quenched disorder: Random couplings

Response from a single bond impurity

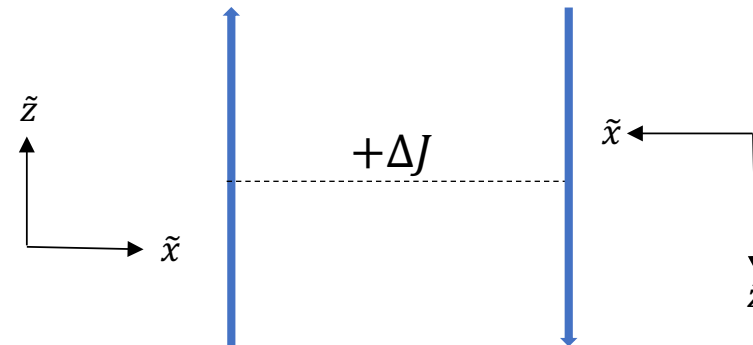
- In a non-collinear magnet



Frustration is locally relieved. For infinitesimal disorder this is a perturbative correction to the bulk Hamiltonian,

$$H \simeq H_0 - h_{\vec{r}}(\tilde{S}_{\vec{r}}^x - \tilde{S}_{\vec{r}+\vec{\delta}}^x)$$

- In a collinear magnet



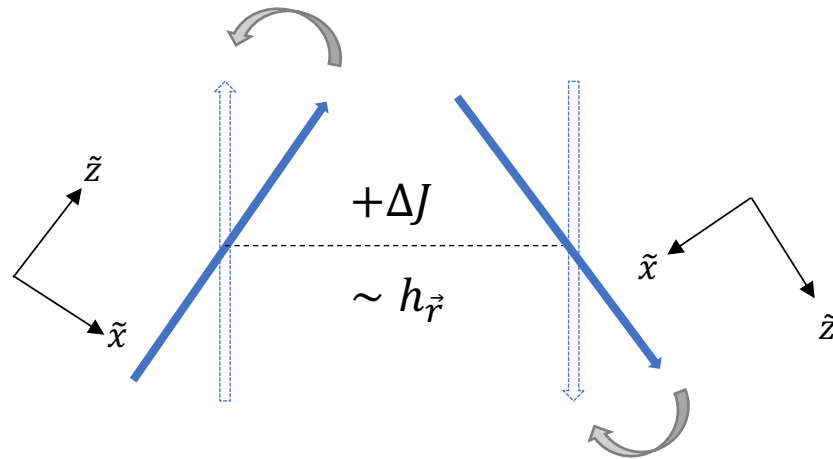
There is no change in the bulk order in response to the bond impurity, hence the Hamiltonian is “virtually” unchanged

$$H \simeq H_0$$

Quenched disorder: Random couplings

Response from a single bond impurity

■ In a non-collinear magnet



Frustration is locally relieved. For infinitesimal disorder this is a perturbative correction to the bulk Hamiltonian,

$$H \simeq H_0 - h_{\vec{r}}(\tilde{S}_{\vec{r}}^x - \tilde{S}_{\vec{r}+\vec{\delta}}^x)$$

- In a linear response approximation this leads to finite expectation value of transverse spin fluctuation.

$$\begin{aligned}\langle \tilde{S}_{\vec{r}}^x \rangle &\sim h_{\vec{r}'} \left(\langle \tilde{S}_{\vec{r}}^x \tilde{S}_{\vec{r}'}^x \rangle - \langle \tilde{S}_{\vec{r}}^x \tilde{S}_{\vec{r}'+\vec{\delta}}^x \rangle \right) \\ &\sim h_{\vec{r}'} \chi_{\vec{r}\vec{r}'}\end{aligned}$$

$$\langle \tilde{S}_{\vec{q}}^x \rangle \sim h_{-\vec{q}} \left(1 - e^{-i\vec{q} \cdot \vec{\delta}} \right) \langle \tilde{S}_{\vec{q}}^x \tilde{S}_{-\vec{q}}^x \rangle \sim h_{-\vec{q}} \chi_{\vec{q}}$$

- If we consider the Goldstone modes, $\langle \tilde{S}_{\vec{q}}^x \tilde{S}_{-\vec{q}}^x \rangle = G(\vec{q}) = \frac{1}{q^2}$, the response is dipolar,

$$\chi_{\vec{q}} \sim (\vec{q} \cdot \vec{\delta}) G(\vec{q}) \sim \frac{\cos \phi_{\vec{q}}}{q}$$

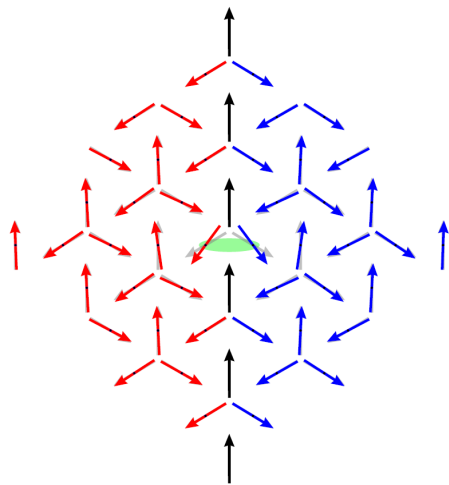
- In 2D, the real space response is also dipolar

$$\chi_{\vec{r}} \sim \frac{\cos \theta_{\vec{r}}}{r}$$

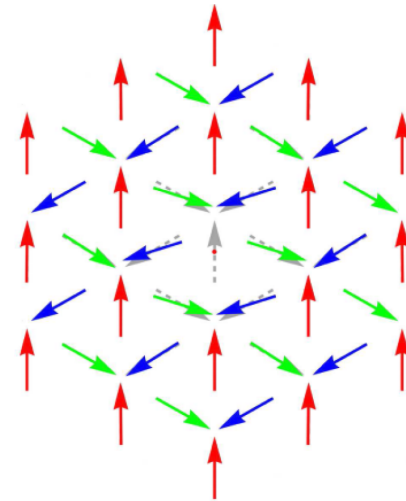
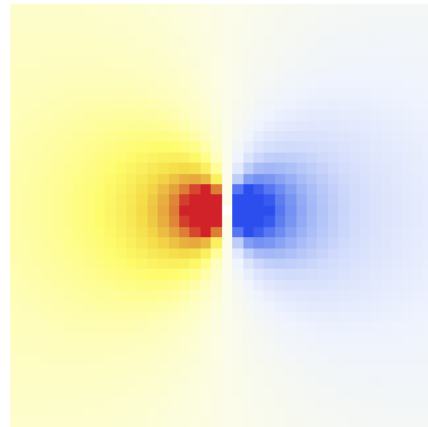
Quenched disorder: Random couplings

Impurities in a 2D non-collinear antiferromagnet

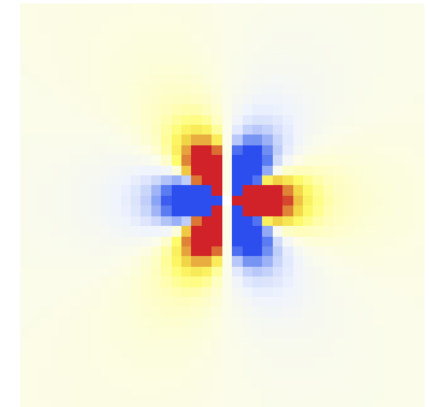
- Bond impurity: Dipolar response
- Vacancy: Octupolar response



$$\delta\theta_r \sim \frac{1}{r}$$



$$\delta\theta_r \sim \frac{1}{r^3}$$



(Wollny, Fritz and Vojta, PRL, 2011)

Quenched disorder: Random couplings

Dipolar response and infrared divergence of frustrated non-collinear magnets

■ Bond impurity: Dipolar response

- Transverse fluctuation is proportional to the strength of the bond impurity.

$$S_{\vec{r}}^{\perp} = \chi_{\vec{r}\vec{r}'} h_{\vec{r}'} = \sum_{\vec{r}'} \chi_{\vec{r}\vec{r}'} h_{\vec{r}'} \quad , \text{ for a distribution of bond impurities.}$$

- For a disordered system, one expects the impurity magnitudes, $h_{\vec{r}}$ to be uncorrelated.

$$\overline{h_{\vec{r}} h_{\vec{r}'}} = h_0^2 \delta^d(\vec{r} - \vec{r}')$$

This implies that the transverse correlation has a divergent contribution,

$$\begin{aligned} \overline{S_{\vec{r}}^{\perp 2}} &= h_0^2 \sum_{\vec{r}'} \chi_{\vec{r}\vec{r}'}^2 = h_0^2 \int \frac{d^d q}{(2\pi)^d} \chi_{\vec{q}}^2 \\ &\sim \int d^d q \left(\frac{1}{q} \right)^2 \end{aligned}$$

- The integral diverges for $d \leq 2$.
- Hence the lower critical dimension for bond disordered non-collinear magnets is,

$$d \leq 2$$

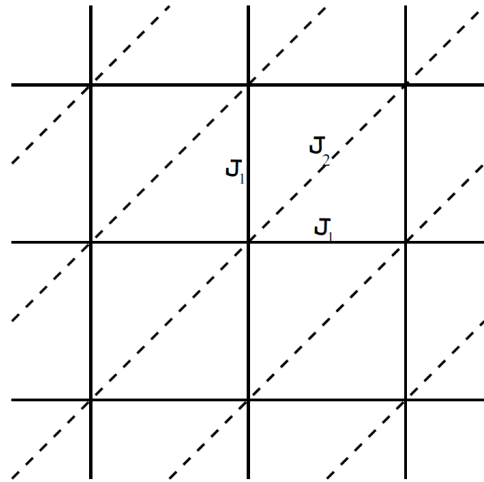
Quenched Disorder: Random Couplings

Lower critical dimension of disordered frustrated non-collinear magnets:
Energetics

Quenched disorder: Random couplings

Disorder energetics: Incommensurability domains

- Incommensurate order in non-collinear magnets
 - Our test system: The triangular lattice Heisenberg model



- In the “isotropic” limit the ground state has spiral order with ordering vector,

$$Q = \left(\frac{2\pi}{3}, \frac{2\pi}{3} \right)$$

- With bond anisotropy the order is described by an incommensurate ordering vector,

$$Q = \left(\cos^{-1} - \frac{J_1}{2J_2}, \cos^{-1} - \frac{J_1}{2J_2} \right)$$

- Claim: In a system with random couplings, some rare regions emerge with a particular bond anisotropy orientation.

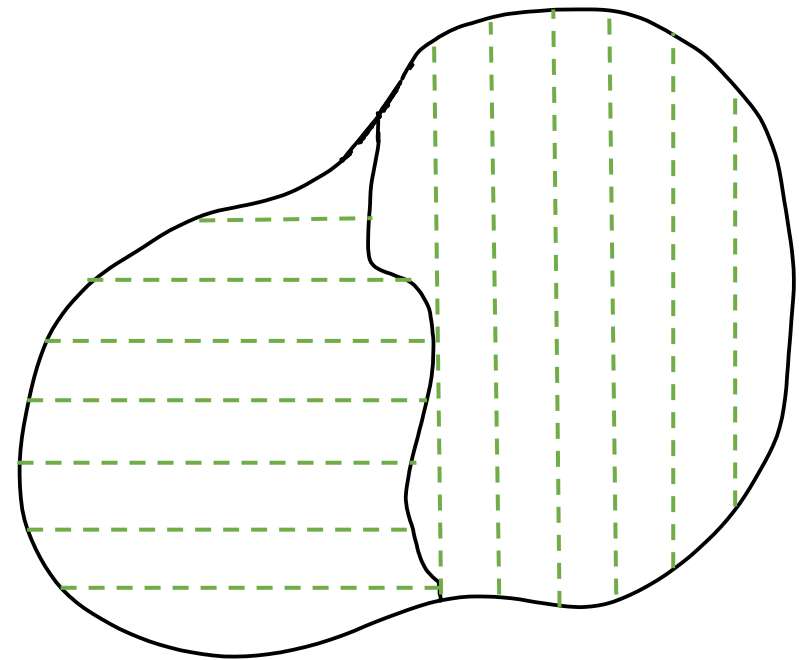
Quenched disorder: Random couplings

Disorder energetics: Incommensurability domains

■ Incommensurability domains:

- In a lattice there is a discrete number of possible bond anisotropy orientation.
- They define a sharp domain wall
- The incommensurate ordering vectors follow the suit
- The surface energy cost for such a domain formation is $\Delta E_{surface} \sim L^{d-1}$, while the bulk energy gain is determined by the volume and the central limit theorem, $\Delta E_{bulk} \sim L^{\frac{d}{2}}$. Hence the critical dimension is once again,

$$d \leq 2$$

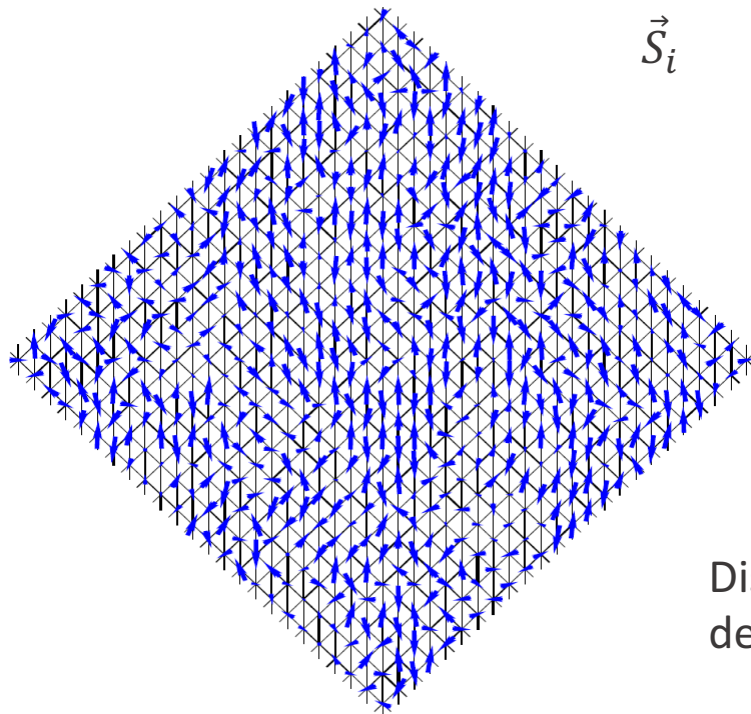


Quenched disorder: Random couplings

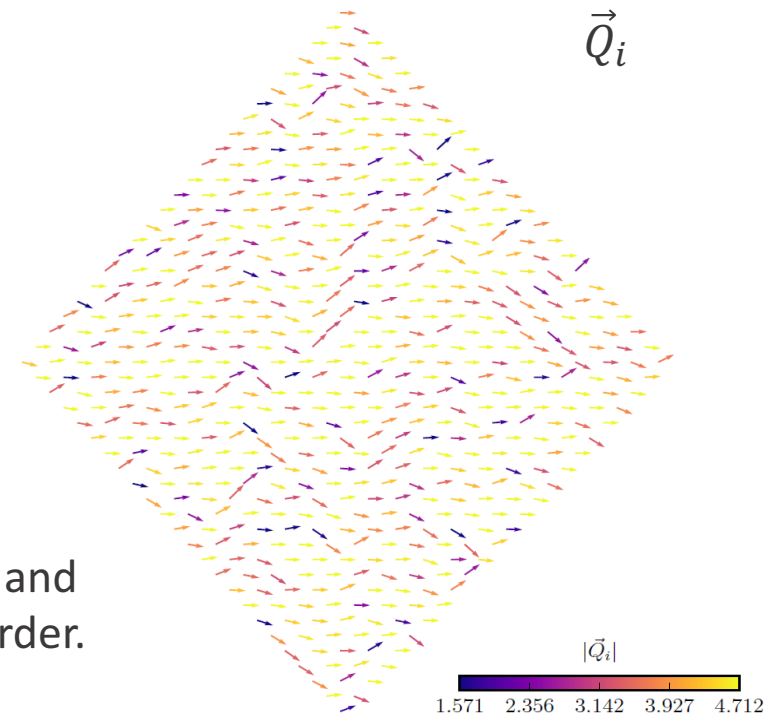
Disorder energetics: Incommensurability domains

■ Incommensurability domains:

- Our test system once again: The disordered triangular lattice antiferromagnet



Discrete domain formation and destruction of long range order.



Quenched disorder: Random couplings

The emergent orders at $d = 2$

- Correlation length grows exponentially with inverse disorder strength

$$C_{h_0}(r) \sim h_0^2 \log\left(\frac{L+a}{r+a}\right) \Rightarrow L_c \sim e^{\frac{1}{h_0^2}}$$

Implication: For small distance resolution, an appearance of order.

- At finite temperature the system is thermally disordered with bare thermal

correlation, $C_T(r) = \frac{1}{\beta} \sum_{\omega_n} \int \frac{d^d q}{(2\pi)^d} G(\omega_n, \vec{q}) e^{i\vec{q} \cdot \vec{r}}$ also diverging. The domination of quenched vs thermal disorder is determined by the ratio,

$$\frac{C_{h_0}(r)}{C_T(r)} \sim \frac{h_0^2}{T}$$

Quenched disorder: Random couplings

The emergent orders at $d = 2$

- In a quasi-2D system with multiple layers, the model essentially has an extra dimension. So, in such a system long range order is not broken as long as layer couplings are finite.
- For a model with SU(2) symmetry broken, if all the low energy modes get gapped out, say, with an energy scale Δ . In effect, long range order survives if,

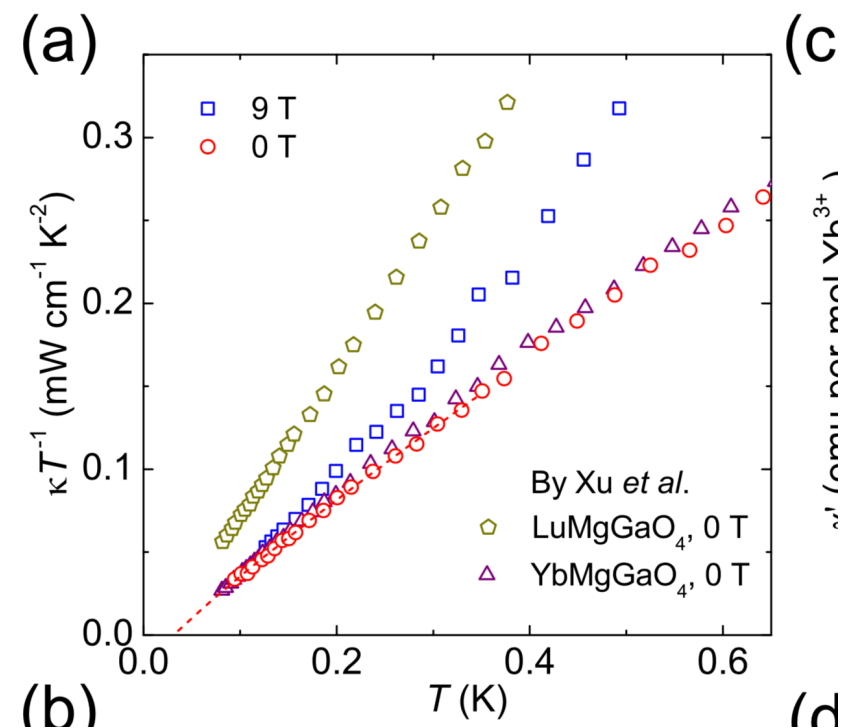
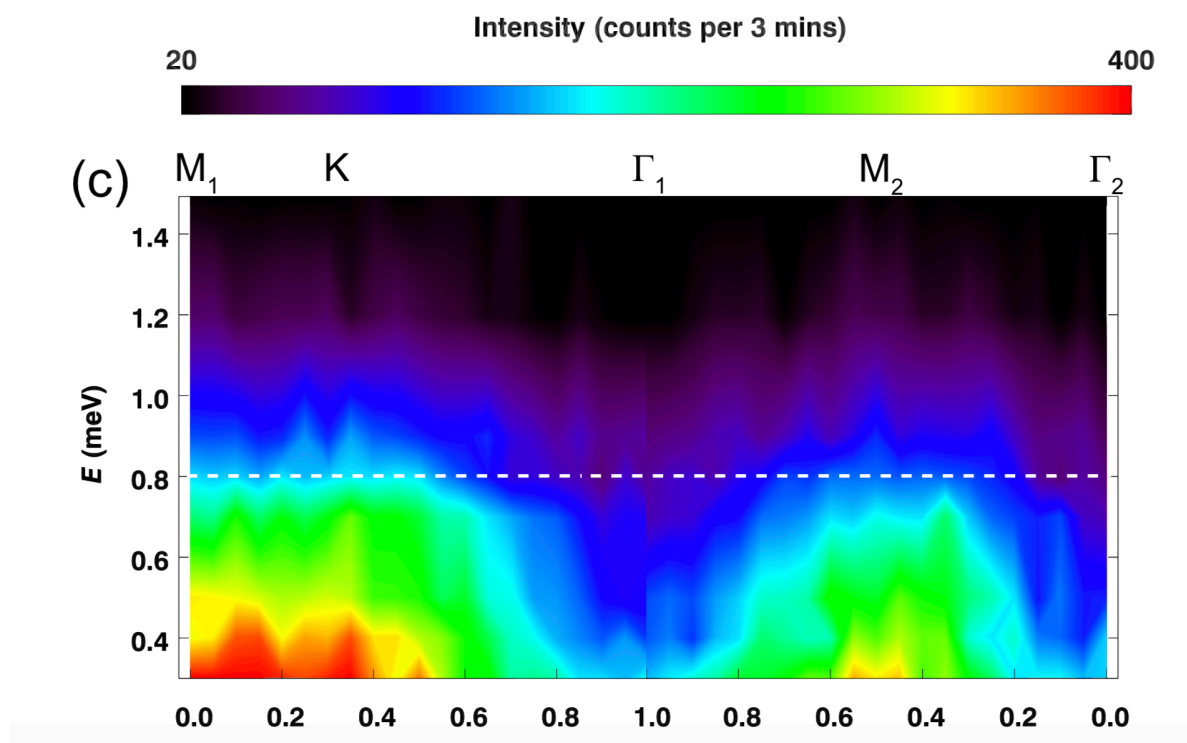
$$h_0^2 \ll \Delta^{2-d}$$

An example

Experiments and microscopics on disordered triangular lattice
antiferromagnet

Disordered Triangular Lattice

Experiments: $\text{YbZnGaO}_4^\dagger$



Thermal conductivity, $\frac{\kappa}{T} = \frac{\kappa_0}{T} + n T^{\beta-1}$

[†]Ma et. al, arxiv, 2017

Disordered Triangular Lattice

Elaborate

Experiments: $YbMgGaO_4$

- A general consensus is that disordered triangular lattice compound is spin-glass in nature.
- “In this Letter, we report comprehensive measurements on a closely related system, $YbZnGaO_4$. We show that the most natural conclusion, that is consistent with the micro- and macro-scopic data presented here is that the system is a spin glass. We believe disorder[5, 8, 11, 16, 17] and frustration[4, 5, 11, 18–22] to be largely responsible for this phase.. Ma et. al”
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- Y. Li, D. Adroja, R. I. Bewley, D. Voneshen, A. A. Tsirlin, P. Gegenwart, and Q. Zhang, Phys. Rev. Lett. 118, 107202 (2017).
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Disordered Triangular Lattice

Quantum fluctuations in the disordered Heisenberg model

- Disordered $J_1 - J_2$ Heisenberg model

SWT

1. Classical ground state with local minimization
2. SWT on top with angle correction

Mean field theory

1. Captures the singlet physics

Disordered Triangular Lattice

Quantum fluctuations in the disordered Heisenberg model

- Signature of disorder
 - A comparison between the spin-wave spectrum and the experimental plot
 - Also comparison between BZ structure factor
 - Mobility edge?
- Spin glass with singlet valleys: pictures from mean field theory
- Qualitative phase diagram

Thank you?

Or a summary?